

Ex.no:9	MINI PROJECT – DIGITAL PLATFORM ANALYTICS: EXPLORING USER INTERACTION AND SOCIAL MEDIA BEHAVIOR

Domain and Problem Statement:

Domain: Social Media Data Analytics

Problem Statement: Social media platforms generate large volumes of data through user posts, likes, comments, shares, saves, views, and other interactions. Analysing this data can help identify engagement patterns, platform performance, content effectiveness, audience sentiment, and influencer behaviour. This project addresses these challenges by transforming raw social media records into structured business intelligence using Power BI and Tableau to evaluate platform-wise engagement, content-type performance, sentiment patterns, influencer-tier engagement, monthly trends, and individual post performance.

Dataset Description:

Dataset Name: Social Media Engagement Dataset (`SocialMediaPosts`).

Granularity: Each row represents a single social media post and its associated engagement and content attributes.

Timeframe Covered: Based on the timestamp/date information available in the dataset.

Key Entities & Attributes:

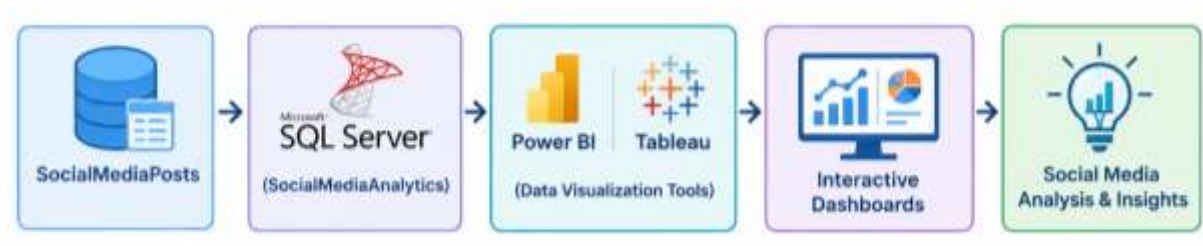
- **Post Identification:** Post_ID, Timestamp
- **Platform Information:** Platform
- **Content Information:** Content_Type, Category, Hashtag_Count, Content_Length, Has_Media
- **Engagement Metrics:** Likes, Comments, Shares, Saves, Views, Engagement_Rate
- **Influencer Information:** Influencer_Tier, Follower_Count, Is_Verified
- **Audience Information:** Sentiment
- **Time Information:** Hour_of_Day, Day_of_Week

Data Source:

Primary Source: Kaggle – Social Media Engagement Dataset.

Storage Infrastructure: The social media records are stored in a local Microsoft SQL Server database named `SocialMediaAnalytics`. The dataset is available through the `SocialMediaPosts` table and is connected independently to Power BI and Tableau for analysis.

Architecture:



Program:

1. SQL Server – Data Retrieval & Preparation

SQL Server Command Prompt:

```
docker ps
docker start sqlserver
```

SQL Server:

```
CREATE DATABASE SocialMediaAnalytics;
GO
```

```
USE SocialMediaAnalytics;
GO
```

Retrieve Social Media:

```
SELECT *
FROM dbo.SocialMediaPosts;
```

Verify Data:

```
SELECT COUNT(*) AS Total_Rows
FROM dbo.SocialMediaPosts;
```

Check for Duplicate Posts:

```
SELECT Post_ID, COUNT(*) AS Duplicate_Count
FROM dbo.SocialMediaPosts
GROUP BY Post_ID
HAVING COUNT(*) > 1;
```

Engagement Rate Analysis:

```
SELECT
    MIN(Engagement_Rate) AS Minimum_Engagement_Rate,
    MAX(Engagement_Rate) AS Maximum_Engagement_Rate,
    AVG(Engagement_Rate) AS Average_Engagement_Rate
FROM dbo.SocialMediaPosts;
```

Post Date Range Analysis:

```
SELECT
    MIN(Timestamp) AS Earliest_Post,
    MAX(Timestamp) AS Latest_Post
FROM dbo.SocialMediaPosts;
```

2. SQL Server – Database and Data Loading

SQL Server:

```
USE SocialMediaAnalytics;  
GO
```

```
SELECT *  
FROM dbo.SocialMediaPosts;
```

Database Details:

Server: localhost\SQLEXPRESS
Database: SocialMediaAnalytics
Table: SocialMediaPosts

Power BI Connection:

Get Data → SQL Server
Server: Server: localhost\SQLEXPRESS
Database: SocialMediaAnalytics
Authentication: Windows
Table: SocialMediaPosts

Tableau Connection:

Connect → Microsoft SQL Server
Server: localhost\SQLEXPRESS
Database: SocialMediaAnalytics
Table: SocialMediaPosts

Procedure:

Step 1: Data Preparation & Preprocessing (SQL Server)

- Open the SQL Server database containing the SocialMediaPosts table.
- Verify the social media records and required fields.
- Check fields such as Post_ID, Timestamp, Platform, Content_Type, and Category.
- Verify engagement fields including Likes, Comments, Shares, Saves, Views, and Engagement_Rate.
- Verify creator and audience-related fields including Influencer_Tier, Follower_Count, Is_Verified, and Sentiment.
- Execute SQL queries to retrieve and validate the required social media data.
- Check the total number of records, duplicate Post_ID values, NULL values, engagement-rate statistics, and post date range.
- Use the validated SocialMediaPosts table as the common data source for Power BI and Tableau.

```

import pandas as pd

# Load dataset
df = pd.read_csv("SocialMediaPosts.csv")

# Create a copy for processing
social_media = df.copy()

# Check missing values
print(social_media.isnull().sum())

# Remove duplicate records
social_media.drop_duplicates(inplace=True)

# Convert timestamp to datetime
social_media["Timestamp"] = pd.to_datetime(
    social_media["Timestamp"],
    errors="coerce"
)

# Display original and cleaned dataset shape
print("Original Shape:", df.shape)
print("Cleaned Shape:", social_media.shape)

```

Figure 1: Social Media Dataset Preprocessing Using Python

social_media.head()

Timestamp	Platform	Content_Type	Category	Likes	Comments	Shares	Views	Saves	Follower_Count	Engagement_Rate	Hour_of_Day	Day_of_Week	Hashtag_Count	Content_Length
2024-01-01 01:42:00	Instagram	Carousel	Business	8257	247	51	29502	20	223080	3.85	1	Monday	15	985
2024-01-01 05:05:00	LinkedIn	Document	Health	1711	27	247	24536	139	312547	0.63	5	Monday	9	627
2024-01-01 08:18:00	Instagram	Carousel	Food	1527	181	7	5400	358	220737	0.78	9	Monday	27	78
2024-01-01 10:58:00	Facebook	Video	Sports	535	178	433	85246	740	428935	0.27	10	Monday	17	554
2024-01-01 13:12:00	Instagram	Photo	Fitness	9706	35	118	25782	611	64384	15.31	13	Monday	5	1136

Figure 2: Preview of the Cleaned Social Media Dataset

Step 2: SQL Database Connection & Data Ingestion

- Open Power BI Desktop → Click Get Data → Select SQL Server.
- Enter the **Server** as localhost\SQLEXPRESS and Database Name as SocialMediaAnalytics.
- Select the appropriate SQL Server authentication method and enter the required credentials.
- Select the SocialMediaPosts table and click Load to import the data into Power BI.
- Open Tableau Desktop → Select Microsoft SQL Server.
- Enter the Server as localhost\SQLEXPRESS and Database as SocialMediaAnalytics.
- Enter the SQL Server credentials and select the SocialMediaPosts table.
- Use the retrieved SQL Server data to create dashboards and interactive visualizations in both Power BI and Tableau.

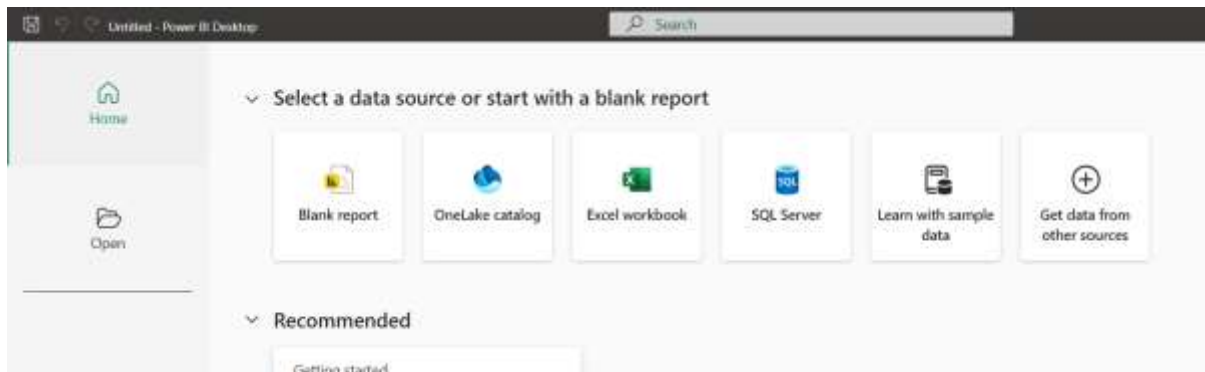


Figure 3: Power BI Desktop Data Source Selection

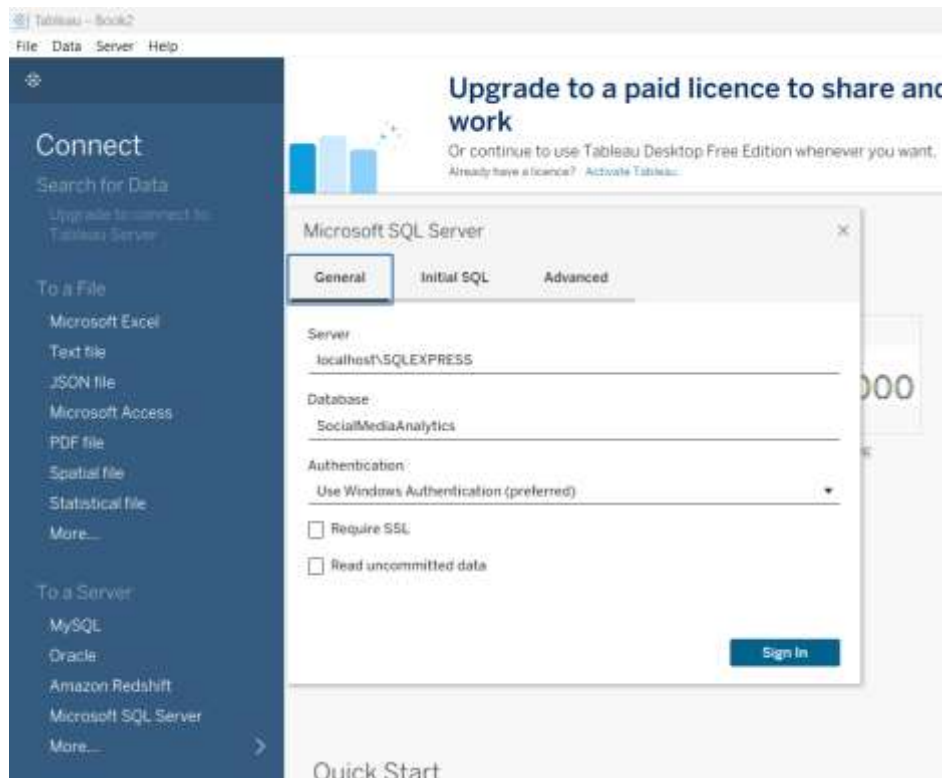


Figure 4: Tableau Connection to SocialMediaAnalytics Database

Step 3: Data Modeling & Relationships (Power BI)

1. Date Table Creation in Power BI:

- Open the imported `SocialMediaPosts` table in Power BI Desktop.
- Create a separate `DataTable` to support time-based analysis.
- Include the `Date` field in the `DataTable`.
- Add date-related attributes such as Month, Month Number, Quarter, and Year.
- Use the `DataTable` fields to perform monthly, quarterly, and yearly analysis of social media engagement.

2. Establishing Relationships in Model View:

- Open the Model View tab in Power BI.
- Verify the `SocialMediaPosts` and `DataTable` tables.
- Establish a relationship between `DataTable[Date]` and the corresponding date field in `SocialMediaPosts`.
- Set `DataTable` on the one (1) side and `SocialMediaPosts` on the many (*) side.
- Ensure that the relationship is active.
- Use the fields from `DataTable` for time-based visualizations and analysis.

3. Applying the Data Model:

- Verify that the relationship between `DataTable` and `SocialMediaPosts` is functioning correctly.
- Use `Month`, `Month Number`, `Quarter`, and `Year` from `DataTable` for time-based dashboard analysis.
- Ensure that the required fields from `SocialMediaPosts` are available for creating measures, calculated columns, visualizations, slicers, and filters.
- Save the Power BI report after completing the data modeling process.

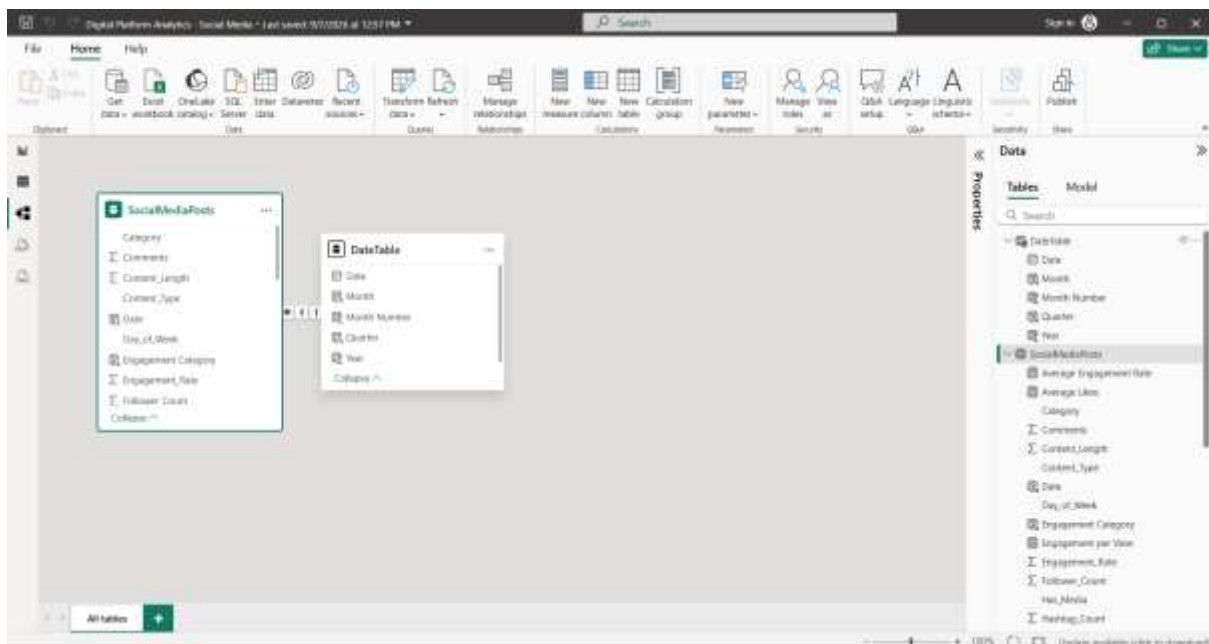


Figure 5: Power BI data model showing the relationship between SocialMediaPosts and DataTable

Step 4: Core Calculations & Measures (DAX)

Power BI Measures (DAX):

The following DAX measures were created to perform key calculations for **Digital Platform Analytics: Exploring User Interaction and Social Media Behavior** and to support the dashboard KPIs and visualizations.

1. Total Engagement

```
Total Engagement =  
    SUM(SocialMediaPosts[Likes]) +  
    SUM(SocialMediaPosts[Comments]) +  
    SUM(SocialMediaPosts[Shares]) +  
    SUM(SocialMediaPosts[Saves])
```

Calculates the total engagement by combining Likes, Comments, Shares, and Saves across the selected social media data.

2. Total Posts

```
Total Posts = COUNTROWS(SocialMediaPosts)
```

Calculates the total number of social media posts available in the dataset.

3. Total Views

```
Total Views =  
    SUM(SocialMediaPosts[Views])
```

Calculates the total number of views recorded across all social media posts.

4. Engagement per View

```
Engagement per View =  
    DIVIDE([Total Engagement], [Total Views], 0)
```

Calculates the engagement generated per view by dividing Total Engagement by Total Views. The `DIVIDE` function returns 0 when the denominator is zero.

5. Average Likes

```
Average Likes =  
    AVERAGE(SocialMediaPosts[Likes])
```

Determines the average number of likes received per social media post.

6. Average Engagement Rate

```
Average Engagement Rate =  
    AVERAGE(SocialMediaPosts[Engagement_Rate])
```

Calculates the average engagement rate across the selected social media posts.

Step 5: Power BI Final Executive Dashboard & Drill-Through

1. Executive Dashboard:

- Place Card Visuals across the top to display key KPIs: Total Posts, Total Engagement, Average Engagement Rate, and Total Views.
- Add a Column Chart for Engagement by Platform using `Platform` on the axis and `Total Engagement` as the value.
- Add a Line Chart for the Monthly Engagement Trend using the date hierarchy (`Year` → `Quarter` → `Month`) on the X-axis and `Total Engagement` as the value.
- Add a Bar Chart for Engagement by Content Type using `Content_Type` on the axis and `Total Engagement` as the value.

- Add Slicers for Platform, Content_Type, Year, Sentiment, and Influencer_Tier to enable interactive filtering of the dashboard.

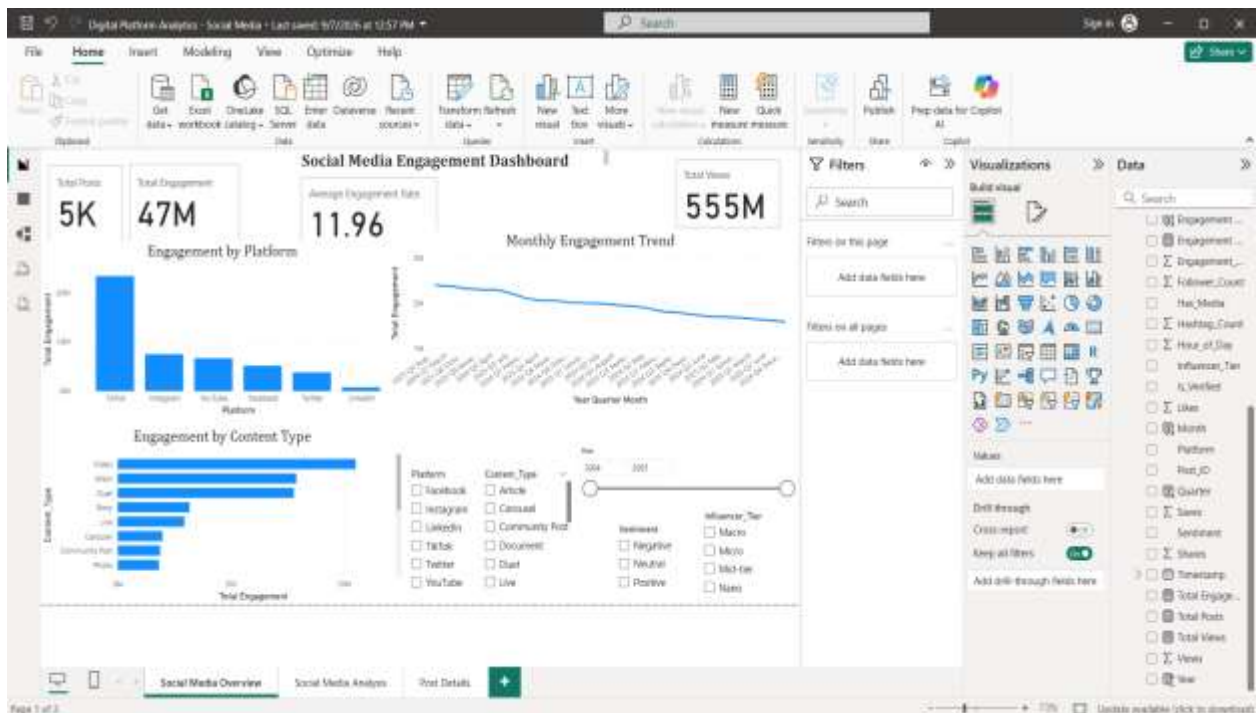


Figure 6: Power BI Executive Dashboard for Social Media Analytics

2. Social Media Analysis:

A separate analysis page was created to provide deeper insights into social media engagement.

- **Engagement by Sentiment** is used to compare engagement across Positive, Neutral, and Negative sentiment categories.
- **Engagement Rate by Platform** is used to compare the average engagement rate across different social media platforms.
- **Likes vs Comments by Platform** is used to examine the relationship between likes and comments across platforms.
- **Engagement by Influencer Tier** is used to compare engagement across different influencer tiers.
- **Top Categories by Engagement** is used to identify content categories generating higher engagement.

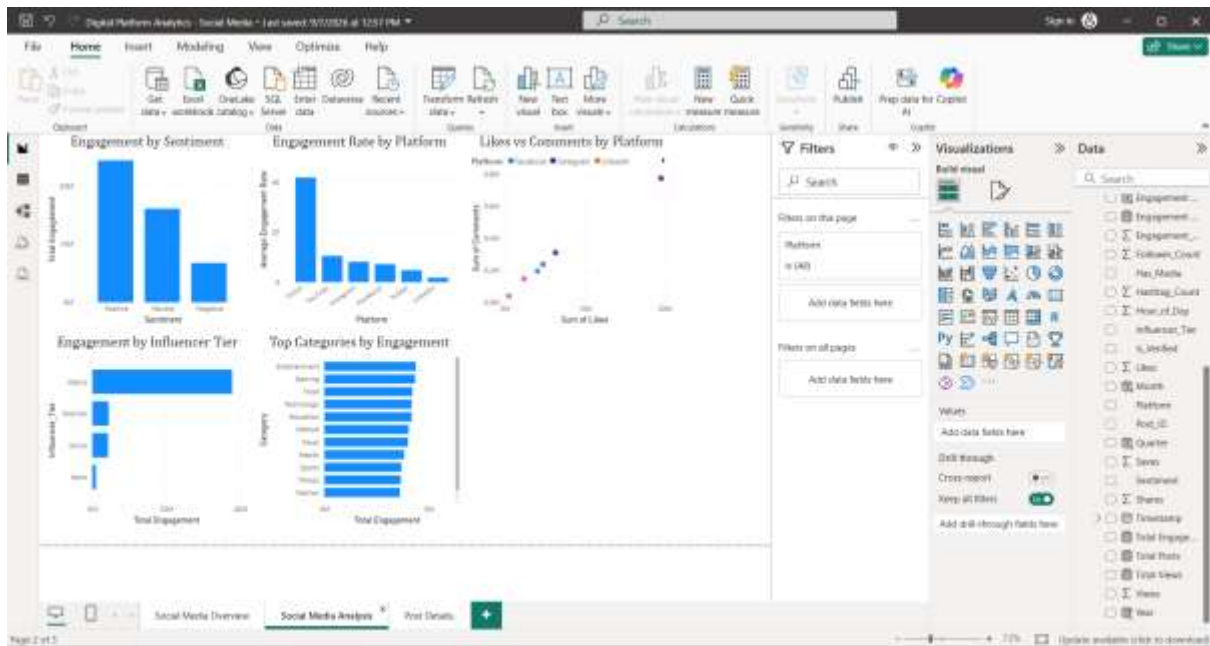


Figure 7: Social Media Analysis Dashboard

3. Post-Level Drill-Through:

A dedicated **Post Details** page was created to provide detailed information for individual social media posts.

The page displays fields such as:

Post_ID, Platform, Content_Type, Category, Likes, Comments, Shares, Views, Engagement_Rate, Sentiment, and Influencer_Tier.

A **Platform drill-through/filter** is applied so that detailed post-level records can be viewed based on the selected platform

Post_ID	Platform	Content_Type	Category	Likes	Comments	Shares	Views	Engagement_Rate	Sentiment	Influencer_Tier
POST_0001	Facebook	Text	Technology	120	15	10	1500	1.2%	Positive	Micro
POST_0002	Instagram	Image	Travel	85	10	5	900	1.1%	Neutral	Macro
POST_0003	Twitter	Text	Business	45	8	3	500	1.0%	Negative	Micro
POST_0004	LinkedIn	Text	Finance	60	12	7	700	1.3%	Positive	Macro
POST_0005	YouTube	Video	Education	30	5	2	400	1.4%	Neutral	Micro
POST_0006	Facebook	Image	Health	90	18	12	1100	1.5%	Positive	Macro
POST_0007	Instagram	Text	Food	110	20	8	1300	1.6%	Positive	Micro
POST_0008	Twitter	Image	Art	70	14	6	800	1.7%	Neutral	Macro
POST_0009	LinkedIn	Text	Science	50	10	4	600	1.8%	Positive	Micro
POST_0010	YouTube	Video	Music	25	4	1	300	1.9%	Neutral	Macro
POST_0011	Facebook	Text	Sports	130	22	15	1600	2.0%	Positive	Micro
POST_0012	Instagram	Image	Fashion	95	16	9	1200	2.1%	Positive	Macro
POST_0013	Twitter	Text	Politics	65	11	5	750	2.2%	Neutral	Micro
POST_0014	LinkedIn	Text	Law	40	7	3	500	2.3%	Positive	Macro
POST_0015	YouTube	Video	Gaming	35	6	2	450	2.4%	Neutral	Micro
POST_0016	Facebook	Image	Real Estate	80	13	7	950	2.5%	Positive	Macro
POST_0017	Instagram	Text	Beauty	100	19	11	1150	2.6%	Positive	Micro
POST_0018	Twitter	Image	Automotive	75	15	8	900	2.7%	Neutral	Macro
POST_0019	LinkedIn	Text	Healthcare	55	9	4	650	2.8%	Positive	Micro
POST_0020	YouTube	Video	Science	30	5	2	400	2.9%	Neutral	Macro

Figure 8: Post-Level Details using Drill-Through

4. Interactive Dashboard Filtering:

The dashboard supports interactive analysis through slicers and visual interactions. For example, selecting **TikTok** from the Platform slicer filters the dashboard visuals to display the corresponding TikTok-related engagement data.

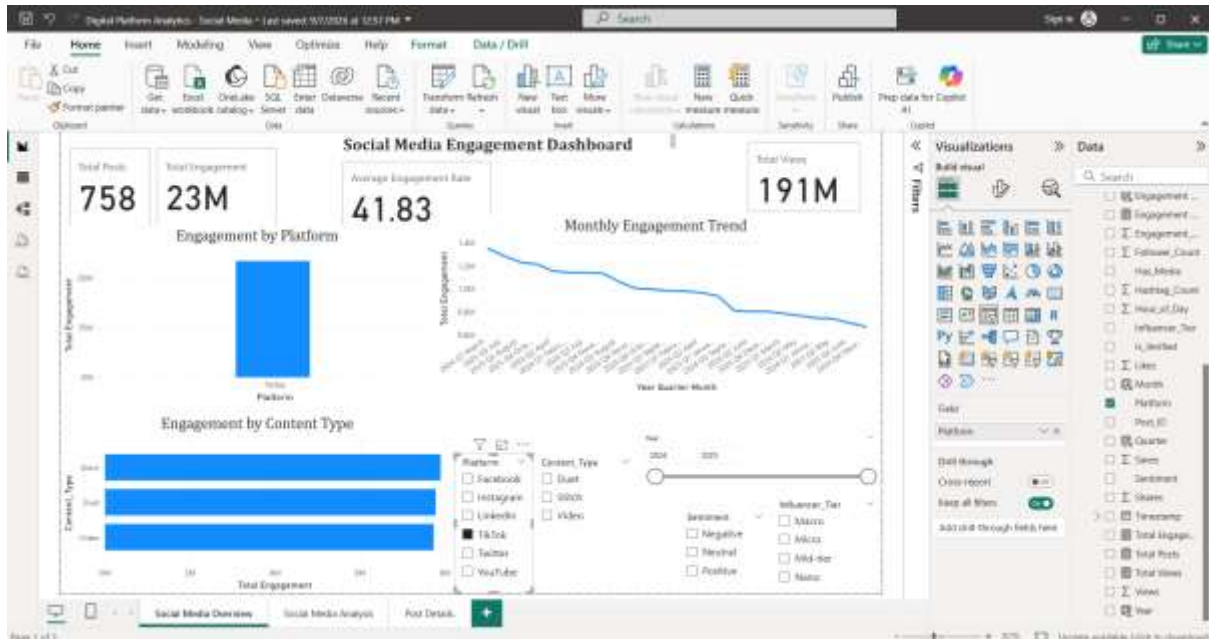


Figure 9: Interactive Power BI dashboard filtered for TikTok platform

Step 6: Tableau Data Connection & Preparation

1. Tableau Data Connection

- Open Tableau Desktop.
- Under Connect, select Microsoft SQL Server.
- Enter the SQL Server connection details:
 - **Server:** localhost\SQLEXPRESS
 - **Database:** SocialMediaAnalytics
 - **Authentication:** Windows Authentication
 - **Table:** dbo.SocialMediaPosts
- Select the `SocialMediaPosts` table to load the social media dataset into Tableau.
- A Live connection was used to connect Tableau directly to the SQL Server database.

2. Tableau Calculated Fields

The following calculated fields/measures were used to support the social media analysis and dashboard visualizations:

Total Engagement

The total engagement measure combines the major engagement indicators such as Likes, Comments, Shares, and Saves.

Engagement Rate

The existing `Engagement Rate` field was used for analysing the engagement performance of social media posts.

Engagement per View

The `Engagement per View` calculation was used to analyse engagement relative to the number of views received by posts.

Likes, Comments and Shares

The `Likes`, `Comments`, and `Shares` measures were used individually for comparative engagement analysis and for the dynamic metric visualization.

3. Creating the Select Metric Parameter

A Tableau parameter named **Select Metric** was created to allow the user to dynamically choose the engagement metric displayed in the dashboard.

The parameter uses **String** data type and provides the following selectable values:

- **Total Engagement**
- **Likes**
- **Comments**
- **Shares**

The default value is **Total Engagement**.

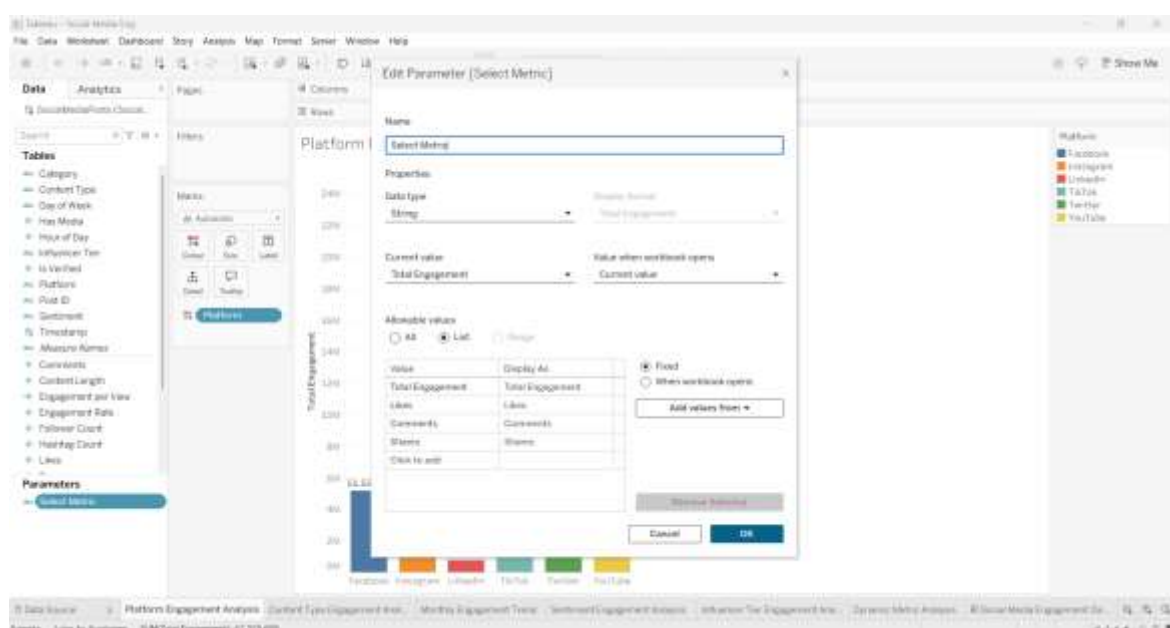


Figure 10: Tableau Select Metric parameter for dynamic metric selection

4. Creating the Selected Metric Calculated Field

A calculated field named **Selected Metric** was created to work with the `Select Metric` parameter. It dynamically changes the metric displayed in the visualization based on the user's selection.

This calculated field is then used in the **Dynamic Metric Analysis** worksheet.

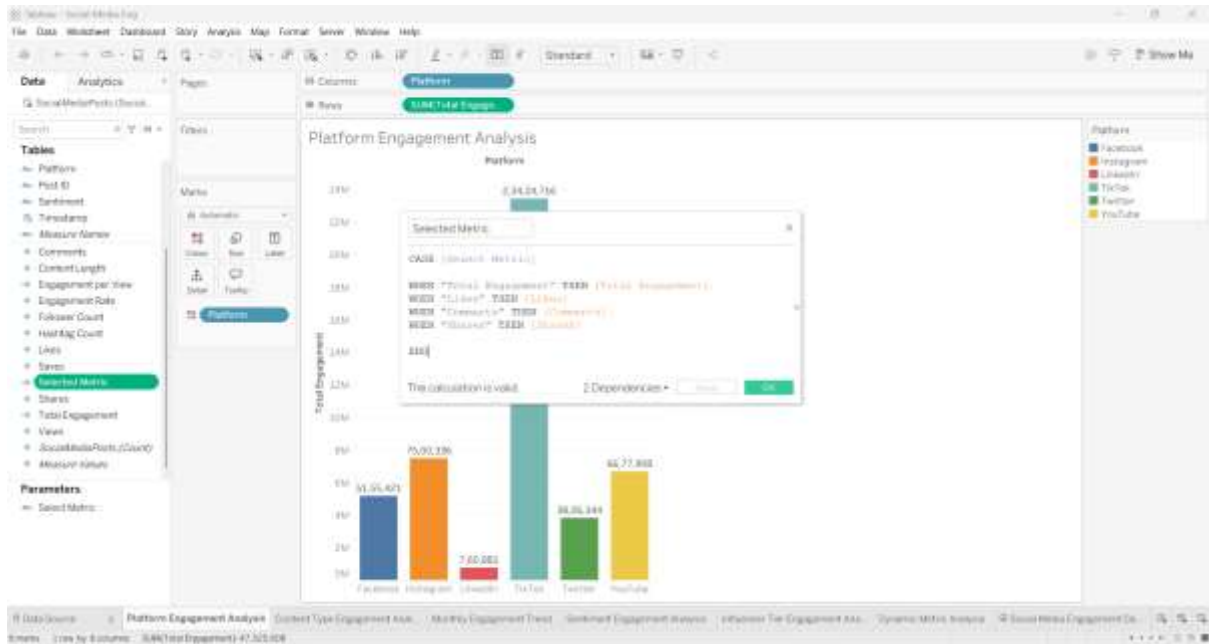


Figure 11: Selected Metric calculated field used for dynamic analysis

Step 7: Tableau Companion Dashboard & Interactive Analysis

1. Tableau Dashboard:

A Tableau dashboard was created to provide an interactive view of social media engagement and content performance.

The dashboard includes the following worksheets and visualizations:

- **KPI – Total Engagement** to display the overall engagement generated by the social media posts.
- **KPI – Total Posts** to display the total number of posts in the dataset.
- **Monthly Engagement Trend** using the timestamp/date information to analyse changes in total engagement over time.
- **Platform Engagement Analysis** to compare total engagement across platforms such as Facebook, Instagram, LinkedIn, TikTok, Twitter, and YouTube.
- **Content Type Engagement Analysis** to compare engagement across different content types such as Video, Static, Duet, Story, Live, Carousel, Community Post, Photo, and other available content types.

- **Dynamic Metric Analysis** to allow users to compare different engagement metrics using the **Select Metric** parameter.

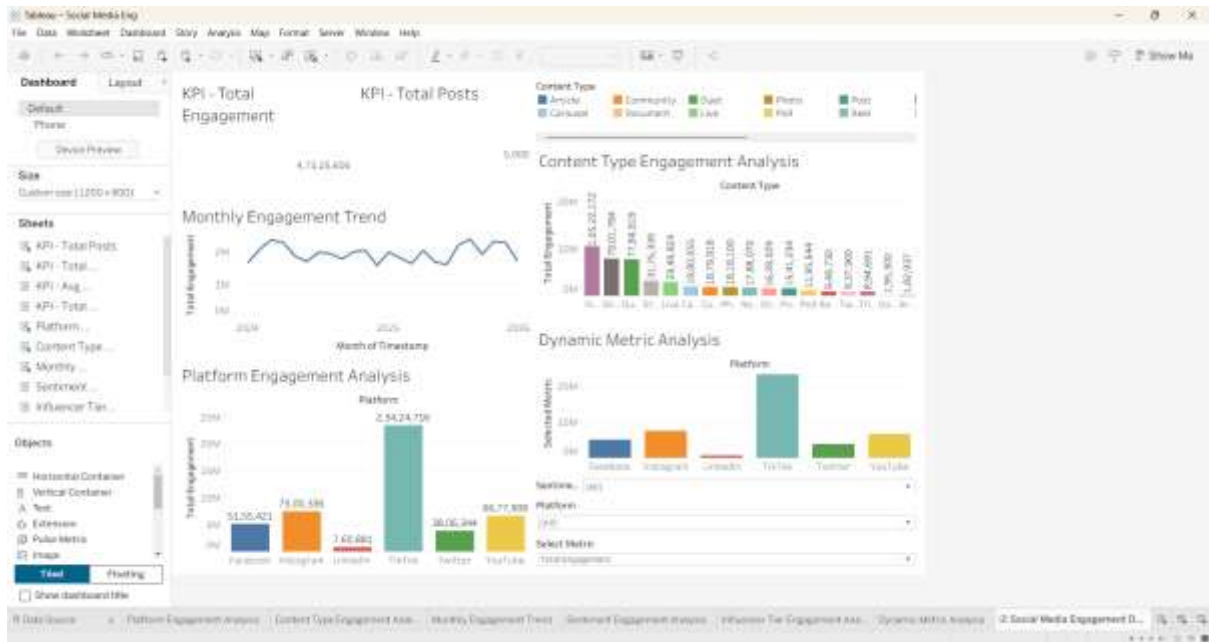


Figure 12: Tableau Social Media Engagement Dashboard

2. Interactive Parameter and Filters:

- A **Select Metric** parameter was created to allow users to dynamically select between Total Engagement, Likes, Comments, and Shares.
- A **Platform** filter was added to allow analysis of individual social media platforms.
- A **Sentiment** filter was added to allow the dashboard to be analysed based on Positive, Neutral, and **Negative** sentiment categories.
- When a different metric is selected from the **Select Metric** parameter, the **Dynamic Metric Analysis** visualization updates accordingly.

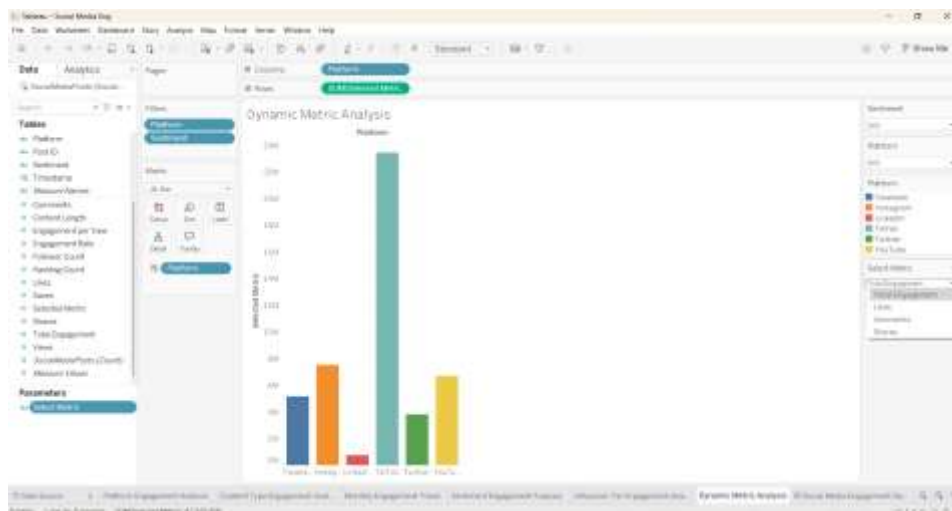


Figure 13: Tableau Select Metric parameter with dynamic metric options

3. Additional Analysis Worksheets:

Additional worksheets were created to support detailed analysis of social media performance, including:

- **Sentiment Engagement Analysis**
- **Influencer Tier Engagement Analysis**
- **Platform Engagement Analysis**
- **Content Type Engagement Analysis**
- **Monthly Engagement Trend**
- **Dynamic Metric Analysis**

These worksheets help analyse engagement patterns across different audience sentiments, influencer tiers, platforms, content types, and time periods.

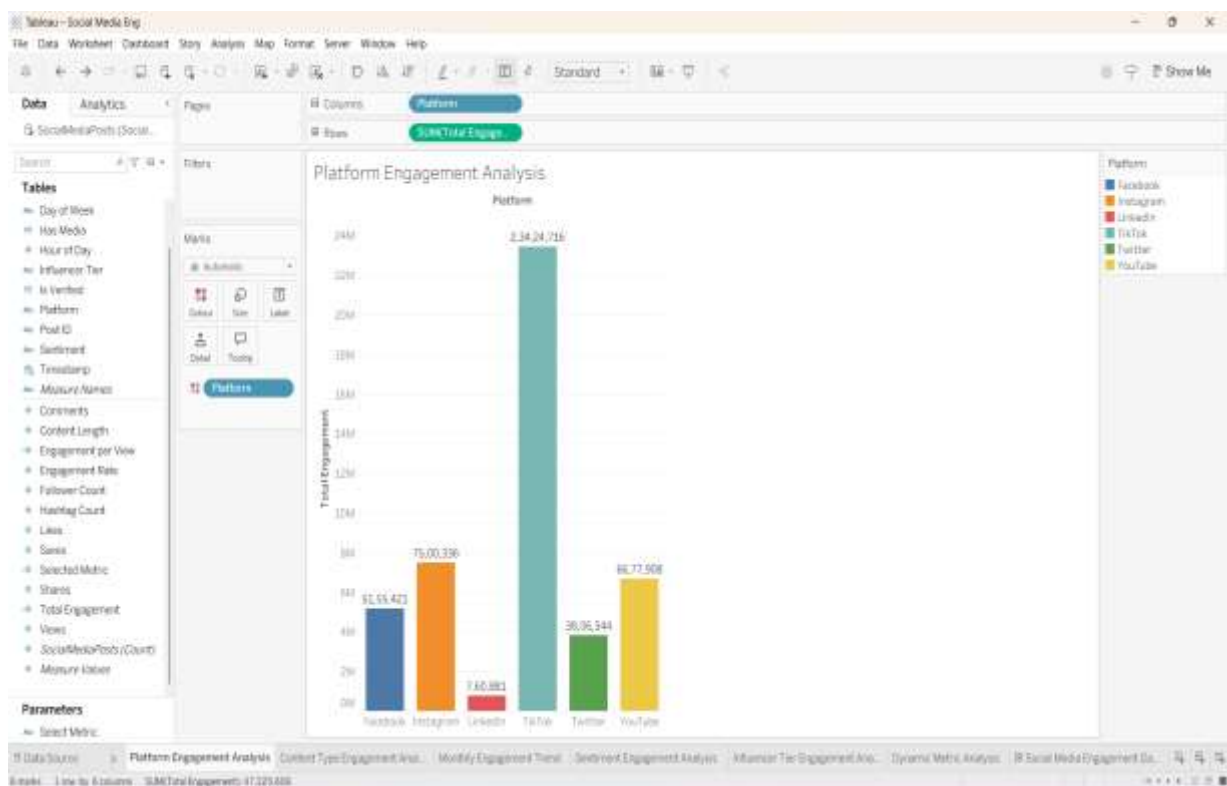


Figure 14: Tableau Platform Engagement Analysis

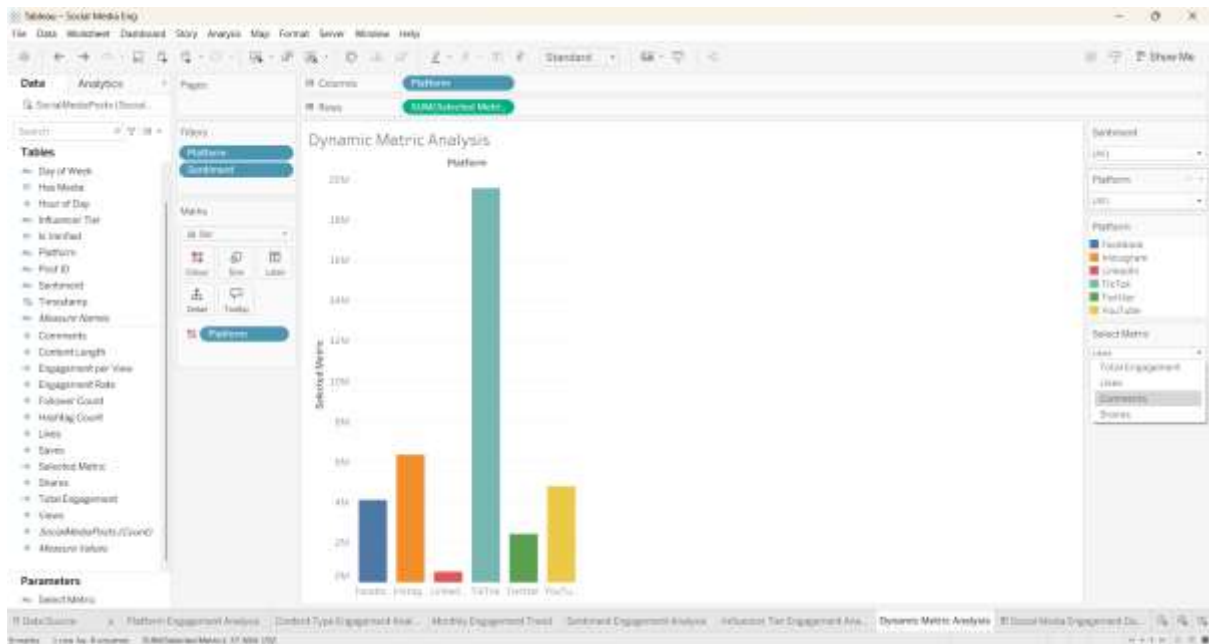


Figure 15: Tableau Dynamic Metric Analysis using the Select Metric parameter

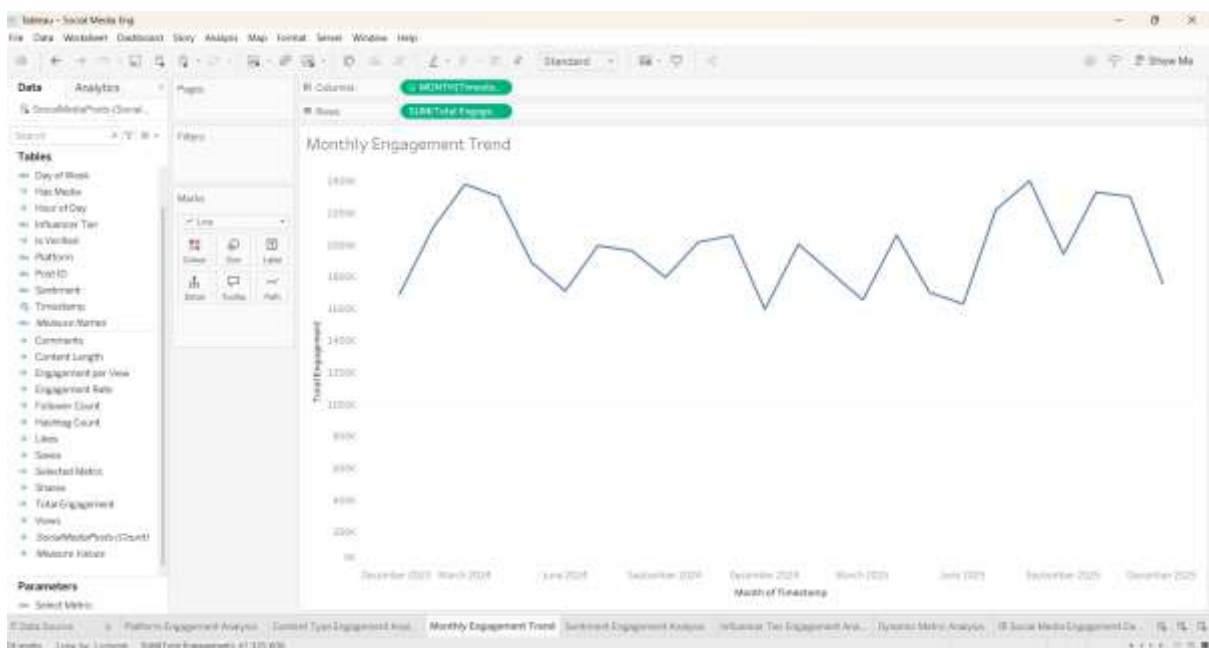


Figure 16: Tableau Monthly Engagement Trend

4. Interactive Dashboard Analysis:

The Tableau dashboard provides interactive exploration through **filters and parameters**. Users can select a specific platform or sentiment and observe the corresponding changes in the dashboard. The **Select Metric** parameter further allows users to switch between Total Engagement, Likes, Comments, and Shares for comparative analysis.

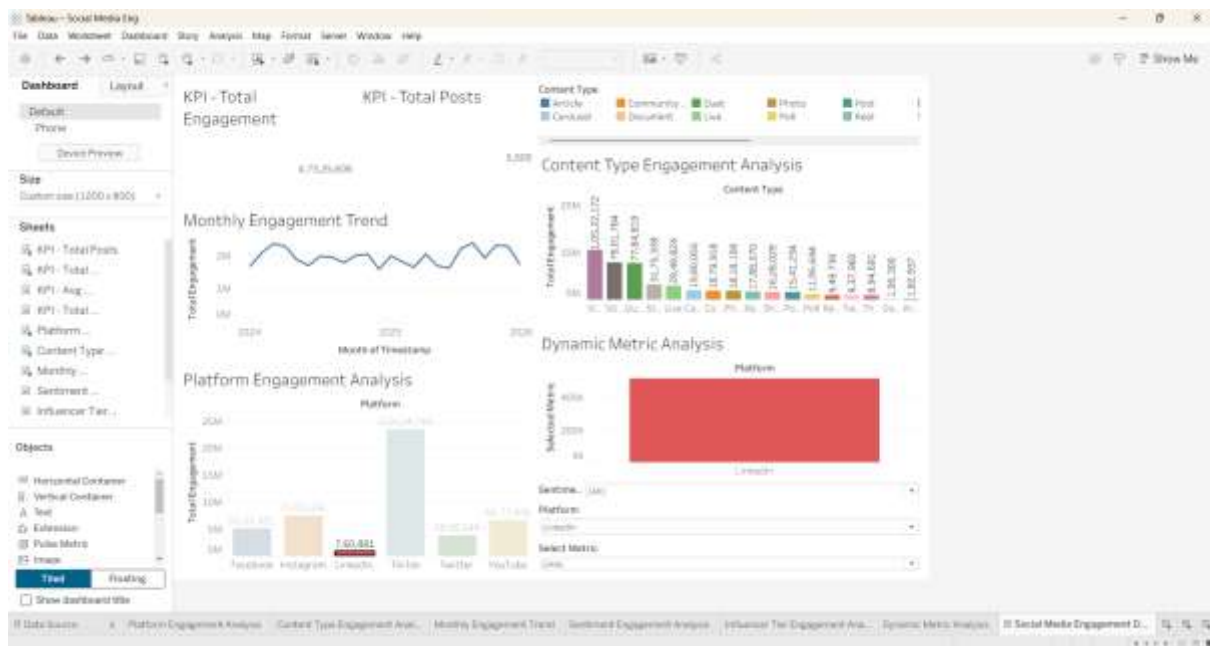


Figure 17: Interactive Dynamic Metric Analysis using the Select Metric parameter